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Data Science

Smart Parking Management System for APSIT

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Project Guide
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1. Introduction

Problem Identified:

- Difficulty in finding available parking spaces for cars and bikes inside the APSIT campus, leading to time wastage and congestion.
- Unorganized parking and vehicles parked in wrong areas due to lack of proper monitoring and information about available slots.

Solution Proposed:

- Develop a Smart Parking Management mobile application that shows real-time parking slot availability through an interactive map.
- Allow users to reserve parking slots, follow designated zones (teacher/student), and receive alerts for wrong parking, ensuring organized and efficient parking management.

2. Objectives

1. To implement a smart parking management system using advanced digital technologies for APSIT.
2. To optimize parking space utilization through a slot-based digital parking system.
3. To enhance parking discipline by implementing slot allocation and restricted parking zones.
4. To integrate real-time monitoring and digital management for efficient campus parking operations.

3. Scope

1. Can ensure efficient parking management in APSIT campus.
2. Can help users quickly find and reserve parking slots.
3. Can be extended to other institutions.
4. Can enable real-time monitoring and improve security.
5. Can minimize time spent searching for parking

4. Feature /Functionality

1. **Feature 1: Secure Login**
   Login using APSIT Moodle ID for authorized access.
2. **Feature 2: Parking Map**
   Shows parking layout with real-time slot availability.
3. **Feature 3: Slot Reservation**
   Reserve parking slots quickly based on vehicle type.
4. **Feature 4: Parking Alerts**
   Notifies users of wrong or restricted parking.

5. Outcome of Project

- User can log in securely using Moodle ID authentication.
- User can view the APSIT parking layout with an interactive parking map.
- User can check real-time availability of car and bike parking slots.
- User can reserve an available parking slot with one tap.
- User can select parking based on vehicle type (car or bike).
- User can identify teacher and student parking zones using color-based segregation.
- User can receive alerts for wrong parking or slot misuse.
- System ensures organized parking, reduced congestion, and better parking management in the APSIT campus.

6. Technology Stack

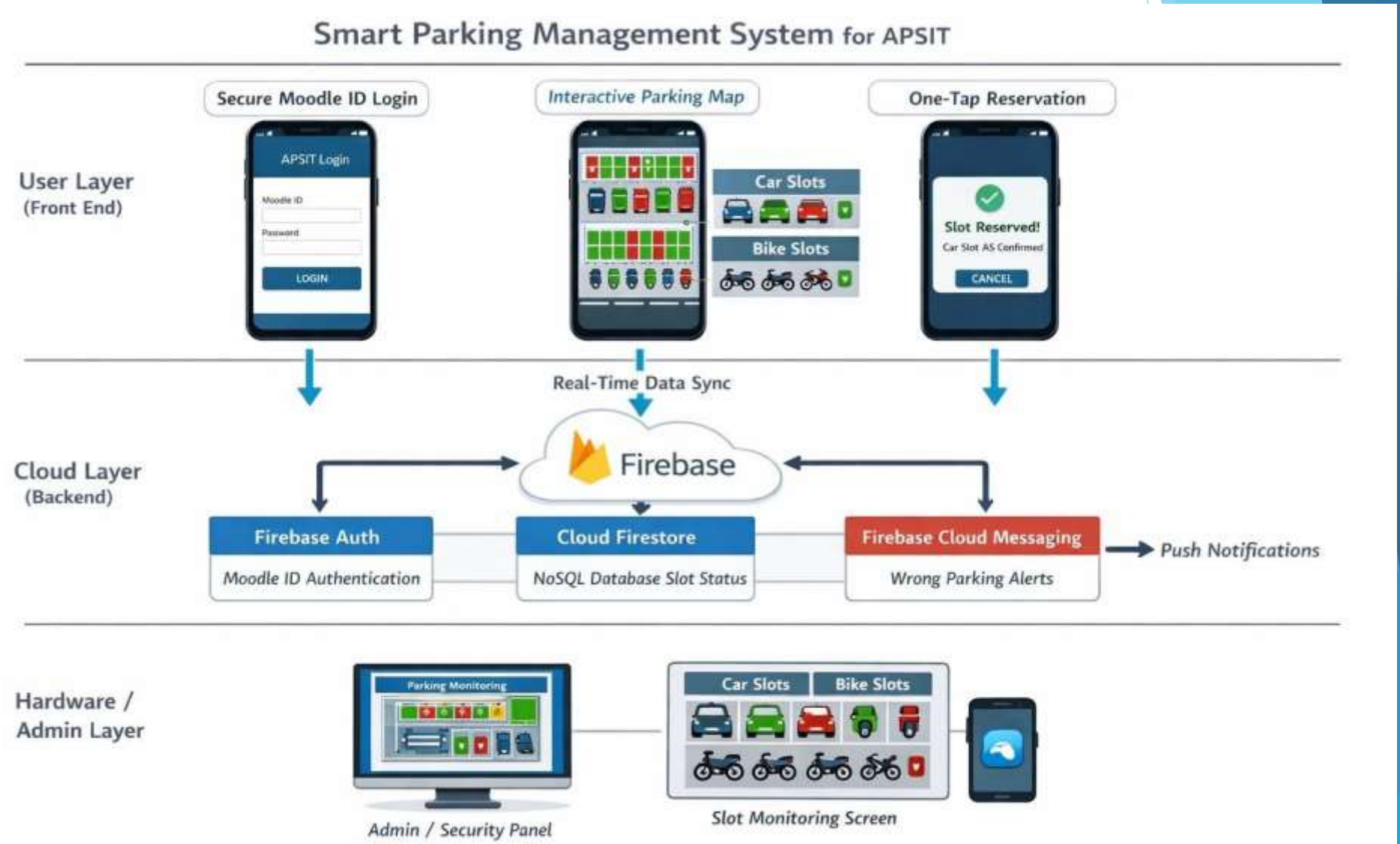
Front End (GUI):-

- **Framework/Language:** Flutter (Dart) for cross-platform Android/iOS.
- **State Management:** Provider for real-time data and auth states.
- **UI:** Custom GridView for parking slots, Google Fonts (Poppins), Material Design 3.

Back End (Database & Services):-

- **Database:** Cloud Firestore (NoSQL) with real-time listeners.
- **Firebase Realtime Database** via APSIT Moodle IDs & staff email/password.
- **Messaging:** FCM for alerts and confirmations.

7. Block Diagram



SDG

SDG 11: Sustainable Cities and Communities

- Reduces traffic congestion
- Promotes organized parking
- Improves safety and efficiency
- Supports sustainable campus management

SDG 9: Industry, Innovation, and Infrastructure

Project uses modern technologies like mobile apps, real-time databases, and digital systems to improve infrastructure within the campus. It promotes innovation by introducing a smart solution for parking management.

APSIT

APSIT Parking

Secure your spot on campus with real-time management.

Student | Teacher

Moodle ID / Email

Enter your ID or email

Password

••••••••

Forgot Password?

Login Now

OR

Login with OTP

Don't have an account? Sign Up

APSIT Parking

• LIVE

Search slot (e.g., A-12)

Car | Bike

ZONE A - MAIN BUILDING 4×3 Layout

A-01	A-02	A-03	A-04
A-05	A-06	A-07	A-08
A-09	A-10	A-11	A-12

SLOT STATUS MEANING

Available · Occupied · Reserved · Faculty

0 Slots Available

Nearby APSIT Wing-C Entrance

Reserve Now

TIME LIMIT 4 Hours

STAFF COST Free

Home | Bookings | History | Settings

My Bookings

Slot A-05 — Active

North Wing - APSIT

MH-04-AB-1234 · 1h · Today, 10:30 AM

View Active Reservation

Slot B-12 — Completed

East Wing - APSIT

MH-04-AB-1234 · 2h · Yesterday, 2:00 PM

Slot A-03 — Expired

Main Building - APSIT

MH-04-AB-1234 · 4h · Feb 22, 9:00 AM

Home | Bookings | History | Settings

Profile

SV

Swati Vishwakarma

Student

24107035@apsit.edu.in

My Vehicle

Edit

MH-04 CS 1813

Royal Enfield Classic 350 • Bike

Account Settings

- Notifications
- Night Mode
- Parking History
- Payment Methods

Logout

APSIT Smart Park v1.0.0

Map | Find Space | Tickets | Profile

Parking History

6 sessions this month

6	13h 15m	₹0
Total Sessions	Total Time	Total Cost

Slot A-05	Feb 24 · 10:30 AM	1h 00m ₹0
Slot B-12	Feb 23 · 2:00 PM	2h 15m ₹0
Slot A-09	Feb 22 · 9:00 AM	3h 45m ₹0
Slot A-03	Feb 21 · 11:00 AM	0h 45m ₹0
Slot B-07	Feb 20 · 8:30 AM	4h 00m ₹0
Slot A-11	Feb 18 · 3:15 PM	1h 30m ₹0

Home | Bookings | History | Settings

Settings

SV **Swati Vishwakarma**
Student

Preferences

- **Push Notifications** — Receive alerts for slot status
- **Location Access** — Used for navigation to slot
- **Auto Release** — Auto-release slot on campus exit
- **Dark Mode** — Switch between light and dark theme

Account

- My Vehicle
- Change Password
- Contact Support
- About App

Logout

APSIT Smart Park v1.0.0

Home | Bookings | History | Settings

Thank You…!!